AgriGuard

AI Software Platform for Autonomous Crop Health Monitoring

Problem

India's farmers increasingly have access to agricultural drones - through ownership or government schemes like Kisan Drone - but these drones are built for spraying and broad-area coverage, not disease detection. Manual field inspection remains the primary way diseases are spotted, which is slow, labor-intensive, and dependent on visual expertise not always available to smallholder farmers. The gap isn't drone or camera hardware - it's intelligence layered on top of the equipment already in the field.

Proposed Solution

We present AgriGuard, an AI software platform that turns any existing drone-mounted camera into a real-time plant disease detection system - no proprietary hardware required. As a drone flies its normal route, captured images are sent to AgriGuard's backend, which works with any camera resolution or aspect ratio without distortion, and classifies 38 disease categories across 14 crop species using an onboard deep-learning model.

Technology

A CNN classifier (trained on the PlantVillage dataset, ~92% validation accuracy) runs via TensorFlow Lite for lightweight, fast inference. An aspect-ratio-preserving preprocessing pipeline ensures accurate results regardless of the source camera's specifications. Detection results feed a live, bilingual (English/Tamil) dashboard showing per-position disease history and trend, an adaptive recommendation for which positions need a closer follow-up pass, automatically generated PDF reports, and a text-to-speech voice summary for farmers who cannot read - alongside soil-moisture-triggered irrigation alerts.

Impact

Because it's pure software layered onto camera-equipped drones farmers already have, AgriGuard requires no new hardware investment, making it deployable at near-zero marginal cost across existing government drone infrastructure. Its camera-agnostic design, adaptive detection logic, and accessibility-first reporting make disease monitoring genuinely usable by the smallholder farmers who need it most.